

Weizheng Wang

COMPUTER VISION · MULTIMODAL LEARNING · UAV/EDGE AI

Adelaide, Australia | Available in China from Sep. 2026

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Research Profile

AI and Machine Learning master's student at the University of Adelaide (expected Sep. 2026), with a Computer Science background and research spanning **multimodal UAV decision evaluation, visual memory, efficient computer vision and edge deployment**. Co-first-author work covers protocol-conditioned benchmarks, maritime small-object detection, structured pruning and detector feature analysis. Engineering experience includes MindSpore model integration, video analytics, Jetson inference, and open-source developer and desktop tools.

Education

The University of Adelaide

Adelaide, Australia

MASTER OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Sep. 2025 – Sep. 2026 (Expected)

- Admitted to the Master of Artificial Intelligence and Machine Learning program with research interests in computer vision, multimodal learning and NLP for UAV and edge intelligence.
- Recipient of the 2025 University of Adelaide Global Citizens 30% International Scholarship.

North China Institute of Science and Technology

Langfang, China

BACHELOR'S DEGREE IN COMPUTER SCIENCE AND TECHNOLOGY

Sep. 2021 – Jul. 2025

- GPA: 90.84/100.
- Coursework and research focused on deep learning, computer vision, and UAV-based edge deployment.

Research Experience

Zayed University

Remote

RESEARCH ASSISTANT – MULROBBENCH: SAFE MULTIMODAL UAV AGENTS

Feb. 2026 – Present

- Co-defined protocol-conditioned UAV decision tasks and designed a 3,024-sample benchmark spanning 17 task-taxonomy nodes under visual degradation and noisy operator instructions.
- Built end-to-end data generation, inference and evaluation pipelines for 17 multimodal models, with semantic and structural diagnostics for safety-policy compliance.
- Co-first-authored the resulting manuscript, submitted to Applied Soft Computing; released as arXiv:2607.23870.

Huawei MindSpore Open-Source Community

Remote

RESEARCH INTERN – NLP & VISION-LANGUAGE MODEL FINE-TUNING

Feb. 2025 – Jun. 2025

- Contributed to the MindNLP open-source project by implementing and fine-tuning models such as Autoformer, BEiT and ALBERT within the MindSpore framework.
- Focused on model integration, training stability and cross-task evaluation across NLP and vision-language tasks, improving reproducibility and robustness of training pipelines.
- Collaborated with open-source mentors and contributors through code reviews, issues and documentation.

Beijing Elite Intelligence Company & NCIST

Beijing & Langfang, China

RESEARCH INTERN – AUTOMATIC INSPECTION AND RESCUE BASED ON DRONE NEST

Jan. 2024 – Jul. 2024

- Built drone-based automatic inspection services in open-pit mines, including water count detection, conveyor belt deviation detection and truck tracking using DeepSORT + YOLOv8.
- Optimized PyTorch detection models and exported them via ONNX for real-time inference on edge devices in long-duration UAV missions.

Hebei Province Key Sci. & Tech. R&D Project (No. 19270318D)

Hebei, China

UNDERGRADUATE RESEARCHER – BAIYANGDIAN ECOLOGICAL IoT 3D MONITORING

Jun. 2023 – Jan. 2024

- Developed object detection for mining trucks using YOLOv7-tiny (Darknet), improving recognition accuracy by 2% with minimal FPS loss.
- Enhanced dual-modal (infrared + visible) UAV object detection with YOLOv8 for low-altitude drone monitoring, achieving 65% mAP at 30 FPS on a laptop and targeting deployment on edge devices.

[†]Equal contribution / co-first authors. All six manuscripts below are submitted, not accepted.

MulRobBench: A Decision-Level Benchmark for Safe and Security-Policy-Compliant Multimodal UAV Agents

Belal S. Alsinglawi[†], **Weizheng Wang**[†], Junyi Wu[†], Lianhai Lin, Mérouane Debbah, Izzat Alsmadi

2026 | *Submitted to Applied Soft Computing* | [arXiv:2607.23870](#)

Co-developed an offline, protocol-conditioned UAV decision benchmark with 3,024 samples, 17 task-taxonomy nodes and 12 scoring dimensions; evaluated 17 multimodal models. Separates semantic decision quality from structural compliance, unsafe actions and parsing failures.

H2T-DEIM: Hilbert-Haar Representation with Tri-Modal Multi-Dimensional Pruning for Maritime Small-Object Detection from Uncrewed Aerial Vehicle Imagery

Junyi Wu[†], **Weizheng Wang**[†], Liqin Tian, Qing Tian, Yong Niu, Yi Jiang

2026 | *Submitted to Remote Sensing*

Combines Hilbert-Mamba context modeling, Haar wavelet downsampling and structured pruning across CNN, Transformer and state-space widths. H2T-DEIM-M achieves 51.1% AP on SeaDronesSee and 64.8% on AFO, with 32.1 FPS on Jetson Orin Nano 8GB under the manuscript's end-to-end protocol; contributed methodology, software and validation.

TRIM: Dual-Budget Structured Pruning with Information-Modulated Saliency for CNN, Transformer, and Mamba Vision Models

Weizheng Wang[†], Junyi Wu[†], Qing Tian, Liqin Tian

2026 | *Submitted to IEEE ICASSP 2027* | [\[Repository\]](#)

Uses dependency-closed pruning units, information-modulated Taylor/Fisher saliency, coupled MILP allocation and measured-budget repair. On ImageNet-1K, Top-1 losses are 0.52/0.66/0.58 percentage points for ResNet-50/DeiT-Base/VMamba-V2 Tiny, retaining at most 69.5% of parameters and 69.4% of forward GFLOPs with zero measured budget violation.

When Retrieval Is Not Binding: A Diagnostic Benchmark for Visual Episodic Memory

Weizheng Wang[†], Junyi Wu[†], Liqin Tian, Qing Tian

2026 | *Submitted to IEEE ICASSP 2027* | [\[Repository\]](#)

Designed matched memory-off, wrong-entity and shuffled-binding interventions while fixing queries, candidates, features and memory envelopes. The development cohort contains 48 objects, 384 observations and 192 queries; on 128 memory-visible queries, three of four training-free readers are unchanged by shuffled binding, separating content benefit from episodic binding sensitivity.

ManiSeg: A Semantic-Entropy Driven Manifold Collision Framework for High-Dimensional Feature Evaluation of Detectors

Junyi Wu[†], Zitong Shen[†], **Weizheng Wang**[†], Xiaoan Shi, Yi Jiang, Liqin Tian

2026 | *Submitted to IEEE ICASSP 2027* | [\[Code\]](#)

Co-developed semantic-entropy-driven granulation, geometric manifold-collision-based adaptive three-way decision and a composite Manifold-Semantic Fidelity metric; evaluated detector feature representations on COCO across multiple model variants.

EmergUAV-Bench: Protocol-Conditioned Evaluation of Vision-Language-Action Models for Emergency UAV Decisions

Junyi Wu[†], **Weizheng Wang**[†], Yi Jiang[†], Liqin Tian

2026 | *Submitted to IEEE ICASSP 2027* | [\[Code\]](#)

Co-developed a 1,280-sample benchmark combining RGB, LiDAR and pose observations with six language conditions, six core emergency tasks and 16 decision facets. Evaluated 13 multimodal models; the best protocol score is 0.4165, with distinct leaders in protocol agreement, safe-action accuracy and unsafe-action avoidance.

Peer-Reviewed Publications

A Lightweight Thermal Denoising and Occlusion-Robust Infrared Detection Model for Substation Equipment

JUNYI WU[†], WEIZHENG WANG[†], YONG NIU, XIAOAN SHI, LIQIN TIAN

• PO-YOLO achieved 83.5% mAP50 and 56.9% mAP50:95 on ISED-1K, improving the YOLOv11 baseline by 3.4 and 2.0 points while reducing computation by 6.35%. [†]Equal contribution. [Code]

IEEE ICIP

2026

A lightweight insulator defect detection algorithm based on drone images for power line inspection

JUNYI WU, WEIZHENG WANG[†], WANYAN DU[†], ZHI LIU, LIANHAI LIN, YUCHEN CUI, CHONG WANG, LIQIN TIAN

• LFL-YOLO achieved 97.4% accuracy while reducing parameters by 23.5% and computation by 9.6% versus YOLOv8n. [†]Equal contribution. [Paper]

Engineering Research Express 7, 045355

2025

Real-time monitoring of trucks used in open pit based on aerial video of UAV

WEIZHENG WANG, JUNYI WU, YONGFEI LI

• Developed a real-time UAV-based monitoring system for open-pit mine trucks using aerial video. [Paper]

IEEE RAIC, pp. 130–138

2024

Research on industrial meter readings based on improved YOLOv8

JUNYI WU, WEIZHENG WANG, LEI YANG, XUESONG NI, LIQIN TIAN

• Proposed an improved YOLOv8 approach for robust industrial meter reading. [Paper]

Proc. SPIE 13181, 131815G

2024

Honors & Leadership

2025	Recipient , University of Adelaide Global Citizens 30% International Scholarship	Australia
2024	Principal Investigator , National Undergraduate Innovation Training Program (202411104021)	China
2023	Grand Award , National 3D Digitization Innovation Competition	China
2023	Second Prize , Challenge Cup Science and Technology Invention and Creation Track	China
2023	First Prize , National College Students' Data Analysis Competition	China

Patents

A Dual-Optical Personnel Search and Rescue Positioning Visual System on Drones

CO-INVENTOR

• Invention patent under substantive examination. Publication No. CN 119429233 A.

China National Intellectual Property Administration

Nov. 2024 - Present

Automatic Battery Change Arm for Drones

CO-INVENTOR

• Utility model patent granted. Patent No. ZL 2023 2 3426097.6.

China National Intellectual Property Administration

Dec. 2023

Research Projects

National Undergraduate Innovation and Entrepreneurship Training

Project lead: 202411104021 (2024); participant in collaborating student projects

Collaborating projects: 202511104021, 202511104025, 202611104287

Undergraduate research in UAV perception, automated inspection and edge inference. Collaborating projects are led by Yuxia Gao, Lucan Tang and Yi Jiang, respectively.

National Key R&D: Multimodal Perception and Emergency Decision Research

Research participant | Maritime detection, model compression, visual memory and UAV decision evaluation

2024YFC3016800; 2024YFC3016805

Contributed to H2T-DEIM, TRIM, ManiSeg, the visual episodic-memory diagnostic and EmergUAV-Bench. Work spans model implementation, evaluation pipelines, feature analysis and protocol-conditioned multimodal benchmarking.

Intelligent Mine Inspection and Monitoring

Research participant | UAV video analytics and industrial inspection

2022YFC3004702-4; HNCTR23-05

Participated in research associated with coal-mine disaster monitoring and digital decision systems, and long-distance coal-flow roller/belt monitoring. Related undergraduate work includes mining-truck detection/tracking, standing-water counting and conveyor-belt deviation detection.

Multimodal Airborne Search-and-Rescue Equipment

Research participant | Equipment development and demonstration | Project budget: RMB 800,000

Related regional programs: 202622003010002; 25350802D; 2025011099

Related work includes infrared-visible UAV detection, personnel search/localization and edge inference. Regional participation covers Hebei major science and technology support, Hebei innovation-capacity enhancement and Langfang science and technology support programs.

IoT Research Platforms and Collaborative Research Funds

Research participant | Sensing, visual detection and research software

21567693H; 2017-ZJ-Y21; 3142024038; 3142026024; 262Y5403D

Hebei IoT Monitoring Technology Innovation Center; Qinghai IoT Key Laboratory; Fundamental Research Funds for the Central Universities; Hebei collaborative innovation special project on livelihood security and social safety. The undergraduate Baiyangdian project (19270318D) is detailed in Research Experience.

Additional Projects & Early Work

Vision and Dataset Utilities

2022–2025 | Python, C++, OpenCV, YOLO *[YOLO tools] [VisDrone] [Face recognition]*

Collected YOLO dataset conversion, splitting, evaluation and open-pit inspection scripts; added VisDrone-VID-to-YOLO conversion and an early OpenCV face-recognition example.

Data Collection and Small Applications

2023–2025 | Python, Scrapy, Selenium, MySQL, PyQt5, JavaScript *[Scrapy] [Video] [HandWrite] [Browser utility]*

Implemented a Scrapy/Selenium answer-collection and text-analysis learning project, a YouTube video-link collection script, a configurable handwriting-page image generator, and a press-and-hold video-speed browser utility.

Academic Website and Presentation Template

2024–2026 | Jekyll, CSS, GitHub Pages, LaTeX, Python *[Website] [Beamer]*

Maintained a personal academic website and a Beamer template with a custom color theme and build/watch automation.

Research Software & Applied ML Projects

Jetson VLM Lab

2026 | Python, llama.cpp, GGUF, CUDA, Docker, Jetson [\[Code\]](#)

Built a shared WSL-to-Jetson validation workflow with model configuration, artifact transfer, runtime launchers, a Python inference client and benchmark logging. Validated text/image and single-frame smoke paths for MiniCPM-V and Gemma; documented Q4/Q8 comparisons and memory gates, with camera and sustained-load validation outside the demonstrated scope.

DEIM Jetson Acceleration Bench

2026 | Python, ONNX, TensorRT, FP16 / INT8, NVIDIA ModelOpt [\[Code\]](#)

Built fixed-shape batch-1 benchmarks with correctness gates, latency percentiles, engine size and board-power measurements. Diagnosed a non-contiguous CHW input binding defect; one-trial FP16 runs measured 89.48 FPS (N) and 48.45 FPS (M) at 640 x 640 with GPU-resident inputs. Selective INT8 was slower; full labeled COCO AP and repeated timing trials remain pending.

Video-based Intelligent Alert System

2025 | Django, Flask, YOLO-World / YOLO-E, ZLMediaKit, SQLite, FFmpeg [\[Platform\]](#) [\[Inference service\]](#)

Developed a multi-stream monitoring platform with role-based access, open-vocabulary detection, alert tasks, event history and multimodal question answering. A separate Flask inference service handles people counting, perimeter intrusion, SQLite event records and alarm clips; supports webcam and RTSP sources with live-stream delivery.

Detector Feature Analysis and Granular Computing

2025–2026 | PyTorch, Ultralytics, scikit-learn, UMAP, t-SNE [\[MC3WD-SEGBC\]](#) [\[Clustering\]](#) [\[Visualizer\]](#)

Implemented YOLO feature extraction, evidential and projection heads, entropy-guided granular balls and geometric three-way decision with training, evaluation, ablation and visualization exports. Additional tools cover three granular/rough-set clustering approaches and per-object t-SNE analysis. The research manuscript is listed separately as ManiSeg.

Model Fine-tuning and MindSpore Integration

2024–2025 | Python, PyTorch, MindSpore / MindNLP, Transformers [\[Autoformer\]](#) [\[BEiT\]](#) [\[ALBERT\]](#) [\[BLIP\]](#) [\[MindNLP\]](#)

Developed Autoformer forecasting experiments on tourism_monthly with MASE/sMAPE evaluation, BEiT fine-tuning on CIFAR-10, ALBERT sentiment fine-tuning on IMDB, and BLIP/MindNLP compatibility experiments. The Autoformer repository also links an upstream MindNLP contribution.

PanelEcho: Comic-to-Audio Prototype

2026 | Python, vision-language models, TTS, web reader [\[Code\]](#)

Built a local pipeline from comic-page images to structured scripts, dialogue/narration/SFX audio and panel-synchronized playback. Includes reusable voice casting, an orchestration backend, inference service and QA reader; the public release is a runnable prototype.

Local AI Translation Application

2025 | Python, NLLB-200 [\[Code\]](#)

Implemented multilingual text and document translation with a local NLLB-200 model, system-tray access and a configurable translation bubble.

Technical Skills

Languages

Python; C++, JavaScript/TypeScript, Java; PowerShell; SQL; LaTeX

ML & Evaluation

PyTorch, MindSpore/MindNLP, Transformers; multimodal benchmarks, visual memory, structured pruning

Vision

YOLO/DEIM, OpenCV, DeepSORT; thermal-visible perception, feature analysis, granular computing

Edge & Deployment

ONNX, TensorRT, llama.cpp/GGUF, CUDA, Jetson, Docker, Linux/WSL

Applications &

Tools

Django, Flask, SQLite/MySQL, ZLMediaKit/FFmpeg; browser extensions; Git, CMake, GitHub Pages

DailyPaper

2025–2026 | Python, GitHub Pages [\[Code\]](#)

Built automated AI/ML/CV/NLP paper collection and a searchable, filterable web archive with category-based browsing and links to original papers.

CleanSlateTab

2025–2026 | JavaScript ES modules, Chromium extensions, WebDAV [\[Code\]](#)

Developed configurable search engines and keyboard-accessible suggestions, wallpaper sources, editable drag-and-drop shortcuts, persistent settings and WebDAV synchronization.

SnapPin

2026 | C++, Windows OCR, CMake, MSVC [\[Code\]](#)

Implemented frozen-frame capture, window/region selection, annotation tools with undo/redo, Windows OCR, multiple pinned image windows and tray-resident single-instance operation.

ChatVirtualizer

2026 | TypeScript, Manifest V3, IndexedDB [\[Code\]](#)

Implemented event-driven virtualization of older chat records, lightweight persisted snapshots and on-demand restoration. Preserves native browser find over collapsed history and handles navigation, message editing and replaced scroll containers; verified locally with browser fixtures.

NotebookLM Exporter

2026 | JavaScript, Manifest V3, DOM parsing [\[Code\]](#)

Built conversation export with lazy-history loading, completeness diagnostics and message selection; preserves sources, citations, tables, code blocks and LaTeX-style math in structured Markdown.

Agent Engineering and Research Tooling

2026 | Python, PowerShell, agent skills, evaluation cases [\[APEX\]](#) [\[Research\]](#) [\[Engineering\]](#) [\[Instructions\]](#)

Created reusable skills for semantic-risk review, evidence-based verification, project-document maintenance, results-to-paper writing and maintainable implementation review. Research tooling includes claim/source templates and behavioral evaluation cases.

Light OCI Manager

2026 | Python, Telegram Bot, OCI, Docker [\[Code\]](#)

Implemented instance/status queries, power operations with confirmation, quota inspection, read-only security-list and boot-volume views, and HTTP health endpoints in a containerized service.

FinderBridge

2026 | Java, Android intents, Xiaomi.eu / HyperOS [\[Code\]](#)

Built a no-root adapter from the launcher's GLOBAL_SEARCH intent to Xiaomi Discovery, using an Activity fallback without background services or special permissions.

UnblockLite Native Process Supervisor

2026 | C++, JavaScript, Windows Job Objects, CMake [\[Code\]](#)

Built a desktop-plugin/native-host split for configuration, readiness and managed child-process lifecycle; uses a private Windows Job Object to terminate owned workers when the parent application exits.

WeChat File Deduplication

2026 | Python, xxhash, SQLite, NTFS, PowerShell [\[Code\]](#)

Implemented month-based incremental scanning, content fingerprints, a SQLite hash index and NTFS hard-link replacement to preserve file paths while avoiding duplicate storage; integrates with Windows Task Scheduler.